Voltage dividers & sensors

National 5 · Electronics & control · Worksheet

ENGINEERING WITH MR KELLY

Name _____ Class _____ Date _____

By the end of this worksheet I can…

- ☐ calculate the voltages in a voltage divider
- ☐ rearrange the voltage divider formula to find a resistor value
- ☐ describe how a sensing circuit works using LDR or thermistor resistance and Vout

I DO — Worked example

Read how a full-mark answer is built

A frost alarm has a thermistor at the bottom of a voltage divider. Its resistance rises as the temperature falls.

1 Describe how the circuit switches the alarm on when it gets cold. (3 marks)

Full-mark answer each ✓ is one mark

As the temperature falls, the thermistor's resistance increases ✓. This means the voltage across the thermistor, Vout, increases ✓. When Vout reaches about 0.7 V the transistor switches on and current flows through the alarm ✓.

Why marks are not awarded

"When it gets cold the thermistor detects it and turns on the alarm."

0 marks: no reference to resistance or Vout.

WE DO — Guided practice

Use the starters and word bank to help you

A voltage divider has R1 = 10 kΩ and R2 = 15 kΩ across a 6 V supply.

2 Calculate V2. (2 marks)

Sentence starters

One … is …, which means …

Another … is …, so …

Word bank

Tip: 2 marks = 2 separate, developed points.

YOU DO **On your own** No notes. Check your answers afterwards.

3 Quick check: tick one box for each.

a) Why replace the fixed resistor in a sensing divider with a variable resistor?
☐ To increase the supply voltage to the circuit ☐ To protect the LDR from too much current ☐ To reverse the current through the sensor ☐ To adjust the level at which the circuit switches

b) A thermistor is at the bottom of a divider. As temperature rises, Vout...
☐ Rises ☐ Reverses ☐ Stays the same ☐ Falls

c) A good description of a sensing circuit starts with...
☐ The sensor's resistance changing ☐ The relay switching on the output ☐ The LED lighting up as a warning ☐ The battery supplying the voltage

d) Vs = 9 V, R1 = 1 kΩ, R2 = 2 kΩ. V2 is...
☐ 3 V ☐ 4.5 V ☐ 6 V ☐ 9 V

4 Key terms: write the letter of the matching description next to each term.

1. **Voltage divider rule**	____	**A** V2 = Vs × R2 ÷ (R1 + R2)
2. **V out of a divider**	____	**B** Vout rises when the sensor's resistance rises.
3. **Sensor at the bottom of a divider**	____	**C** V1/V2 = R1/R2
4. **LDR at the bottom**	____	**D** Sensor resistance → Vin → transistor → relay → output.
5. **Describing a sensing circuit**	____	**E** Vout rises as it gets darker (resistance rises).

A thermistor is R2 in a voltage divider with a fixed resistor R1 = 4.7 kΩ and a 5 V supply. At 20 °C the thermistor is 2.2 kΩ.

5 Calculate Vout across the thermistor at 20 °C. (2 marks)

A light-sensing circuit needs V2 = 1.5 V from a 6 V supply. R2 (an LDR) is 1.2 kΩ at the switching light level.

6 Calculate the value of R1. (3 marks)

A voltage divider has a 24 V supply. R2 is 26 Ω and V1 across R1 must be 9 V.

7 Calculate the resistance of R1. (3 marks)

Check and reflect

Mark your answers with the answer sheet. Go back to the checklist on the site and re-rate each “I can” statement above. Which one still needs work?